

TABLE 1
THE STATUS OF MAJOR SEMICONDUCTOR MANUFACTURING PROJECTS IN EUROPE

Company	Location	Investment Amount	Project Details	Expected Completion Date
European Semiconductor Manufacturing Company (ESMC)	Dresden, Germany	€10 billion	TSMC, Bosch, Infineon, and NXP joint venture for automotive and industrial chips. Expected by 2027.	2027
GlobalFoundries	Dresden, Germany	\$8 billion	Expansion to double semiconductor production capacity.	Not specified
Intel Corporation	Magdeburg, Germany	€30 billion	Fab construction pending approval, expected completion in 4-5 years.	4-5 years after approval
STMicroelectronics	Catania, Italy	€292.5 million	Silicon carbide wafer plant supported by EU Recovery Facility. Target completion in 2026.	2026
Photonic Chip Plants	Netherlands	€133 million	EU investing in photonic semiconductor pilot facilities. Development to start in 2025.	2025
Wolfspeed and ZF	Ensdorf, Germany	€3 billion (postponed)	Electric vehicle microchip factory. Project indefinitely postponed due to low demand.	Indefinitely postponed
Intel Ireland Fab 34	Leixlip, Ireland	\$7 billion	Expansion of existing Intel campus for 7nm process technology. Expected to be operational in 2023.	2023
Nexperia	Manchester, UK	Undisclosed	Expansion of wafer fab to increase power semiconductor production capacity.	Not specified
Tower Semiconductor & STMicroelectronics	Agrate Brianza, Italy	€3 billion	Joint venture for a 300mm fab focused on analogue and mixed-signal technologies. Expected by 2026.	2026
X-FAB	Dresden, Germany	€200 million	Upgrading 200mm wafer production for automotive and industrial applications.	Not specified
Bosch Reutlingen Expansion	Reutlingen, Germany	€400 million	Expansion to increase automotive semiconductor production. Expected completion by 2025.	2025
Siltronic AG	Freiberg, Germany	€2 billion	New 300mm wafer plant for semiconductor manufacturers. Operations planned by 2025.	2025
NXP Semiconductors	Eindhoven, Netherlands	€500 million	Facility upgrade to enhance automotive semiconductor production. Expected by 2024.	2024

local chip production and reduce dependence on Asia. Originally planned for electric vehicle (EV) chip production, the project has been pushed back by two years, with construction now expected to begin in mid-2025 as Wolfspeed seeks additional funding amid internal investment pressures and broader market challenges in Europe and the US.

Across the Atlantic, Wolfspeed announced further restructuring, closing its 150mm Durham fab in North Carolina and another facility in Texas, alongside a 20% workforce reduction. The company cited struggles in the EV market as the driving force behind these moves. In 2024, a Cox Automotive report showed US EV sales rose 7.3% to 1,301,411 units, making up 8.1% of total sales (up from 7.8% in 2023). In contrast, European

BEV registrations fell 10.2% in December, with annual volumes down 5.9%—driven by sharp declines in Germany (-38.6%) and France (-20.7%), according to ACEA (European Automobile Manufacturers' Association).

Intel has also been grappling with delays. Its Magdeburg, Germany, fabs—Fab 29.1 and Fab 29.2—have been postponed due to EU subsidy approval delays and soil removal requirements. Originally set to begin in summer 2024, construction will now start in May 2025. The two fabs, intended to house Intel's advanced 14A (1.4nm) and 10A (1nm) manufacturing technologies, were initially expected to start production by late 2027. That timeline has now slipped to between 2029 and 2030.

Intel's Ohio-based chip projects face similar setbacks. The \$20

billion investment in two fabs in Licking County, announced in January 2022, was expected to be operational by 2025. However, weak market demand and delayed government subsidies have pushed construction to 2026-2027, with production now slated for 2027-2028.

On the other side of the coin

In a separate development, three semiconductor projects made headlines recently. German-based Infineon has officially started constructing a new backend semiconductor factory in Thailand. Indian company Indichip, alongside Japan's YMTL, has made a joint investment of \$1.6 billion to establish India's first silicon carbide (SiC) wafer manufacturing facility. Last but not least, Fuji Electric has begun mass production of 6-inch